

DropBox

Functional requirements;

user should be able to upload/download any file based on his preference,

user should be able to search for any file based on his preference,

Non-functional requirements:

Data-partitioning,

load balancing,

fault tolerance,

vertical scalability,

horizontal scalability,

low latency,

high throughput,

CAP theorem (Here we eventual consistency is fine , but here we need 99.99% uptime

i.e Availability>>Consistency)

Core Entiites:

user,

file,

history,

API:

POST:/upload

{

 userID.

 filemetadata

}

GET:/search/{fileID}

HLD:

